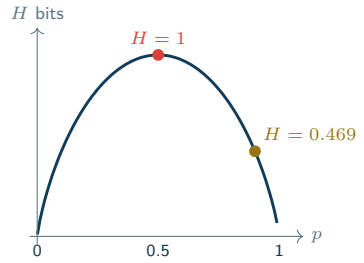
PROBABILITY LADDER 9 OF 9

Information and entropy

The surprise of an event is the logarithm of its improbability, and entropy is the average surprise.

1 / THE RULE AND THE PICTURE

An event of probability p carries $-\log_2 p$ bits of surprise, and the entropy is their average



entropy of a coin that lands heads with probability p

$$H = -\sum_i p_i \log_2 p_i = E[\text{surprise}]$$

A fair coin gives 1 bit, a coin that lands heads 90% of the time gives 0.469 bits because the outcome is usually predictable, and the entropy is largest when all outcomes are equally likely.

2 / SOME ENTROPIES

The more evenly spread the distribution, the more bits it carries

source	entropy	note
fair coin	1 bit	$\log_2 2$
coin with $P(\text{heads}) = 0.9$	0.469 bits	$-0.9\log_2 0.9 - 0.1\log_2 0.1$
fair die	2.585 bits	$\log_2 6$
sum of two dice	3.274 bits	less than $\log_2 11 = 3.459$, since the sums are not equally likely

For W equally likely outcomes the entropy is $\log_2 W$, which is the largest value possible for W outcomes.

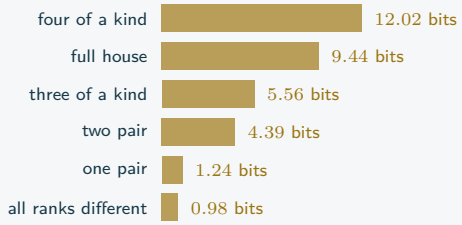
3 / ENTROPY AS THE LENGTH OF A CODE

No code can use fewer bits per symbol on average than the entropy

For four symbols with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$ the codewords $0,\ 10,\ 110,\ 111$ have lengths $1, 2, 3, 3$, so the average length is $\frac{1}{2}\cdot 1 + \frac{1}{4}\cdot 2 + \frac{1}{8}\cdot 3 + \frac{1}{8}\cdot 3 = 1.75$ bits, which equals the entropy because each codeword length is exactly $\log_2(1/p)$. Likely symbols get short codewords and the code needs no separators, which is the idea behind data compression.

4 / A POKER HAND / SURPRISE AND THE NUMBER OF MICROSTATES

A hand pattern with W hands has probability $W/2\,598\,960$ and surprise $\log_2(2\,598\,960/W)$



The six rank patterns have entropy $H = 1.365$ bits, while specifying the exact hand takes $\log_2 \binom{52}{5} = 21.31$ bits, and the difference 19.94 bits is the average $\log_2 W$ left within a pattern. The rare patterns are the surprising ones: a full house has probability 0.00144, a surprise of 9.44 bits.

5 / CONNECTION TO PHYSICS

Boltzmann's entropy is the entropy of a uniform distribution over microstates

pattern	hands W	probability	surprise (bits)	$\log_2 W$
four of a kind	624	0.00024	12.02	9.29
full house	3 744	0.00144	9.44	11.87
three of a kind	54 912	0.02113	5.56	15.74
two pair	123 552	0.04754	4.39	16.91
one pair	1 098 240	0.42257	1.24	20.07
all ranks different	1 317 888	0.50708	0.98	20.33

For a uniform choice among W microstates the entropy is $\ln W = 14.77$ nats for the whole pack of hands, and Boltzmann's $S = k \ln W$ is this quantity in physical units; the surprise of a pattern equals $\log_2(2\,598\,960) - \log_2 W$, so patterns with more hands are less surprising.

ASIDE / BITS AND NATS

Using $\ln$ in place of $\log_2$ changes the unit from bits to nats, with 1 nat $= 1.443$ bits.

LINKS TO OTHER SHEETS

The poker counts and the entropy of the card pack are developed in the card-pack entropy sheet; the previous sheet is *The normal distribution and the central limit theorem*.

Convention. H is the Shannon entropy; $\log_2$ gives bits; W is the number of equally likely hands in a pattern; k is Boltzmann's constant.